Gaspard Nordin

French engineering student at Arts et Métiers Institute of Technology, completing the MSc KIMP research master in Industrial Engineering, with a strong focus on the modelling and simulation of production systems.

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EDUCATION

Arts et Métiers — MSc KIMP — Paris, France **Sep 2026 – Sep 2027**
Master of Science in Industrial Engineering (research master) — Knowledge Integration in Mechanical Production, Design & Manufacturing track
International research master taught in English, backed by the 14 laboratories and research teams of the Arts et Métiers network and designed as a pathway to doctoral research.

- Integrated design & manufacturing, CAD/CAM and geometric modelling, AI for design, mechatronics modelling & control.
- Manufacturing process management, robust design & big data, digital mock-up & virtual reality, decision & risk analysis.

Arts et Métiers Institute of Technology (ENSAM) — Metz & Paris, France **Sep 2024 – Present**
Engineering Degree (Diplôme d'Ingénieur) — Generalist Engineering, Programme Grande École
A founding member of ParisTech, renowned for its excellence in mechanical and industrial engineering, combining fundamental physics with hands-on industrial practice. Pursued in parallel with the MSc KIMP.

- Data Science, Mechanics, Thermodynamics, Materials Science, Supply Chain.
- Exchange semester at Universiti Teknologi PETRONAS, Malaysia (Mar – Aug 2026): oilfield and process-related corrosion, condition-based monitoring and analysis, mechatronics, control systems, building air-conditioning systems.

Lycée Lakanal — Sceaux, France **Sep 2021 – Jul 2024**
Intensive preparatory classes (CPGE), Mathematics & Physics (MP)
Intensive programme preparing for the competitive entrance exams of the French Grandes Écoles d'ingénieurs.

- Around 35 hours of lectures and tutorials per week; a 4-hour written exam and 2 to 3 one-hour oral examinations every week.
- Linear algebra, real & complex analysis, probability, classical mechanics, electromagnetism, thermodynamics, optics & waves.

SELECTED PROJECTS

Research Project — Inclusive Task Allocation **Sep 2025 – Dec 2025**
Team of 5 — Stochastic modelling of a production system (phase 1)
Objective : Model a workshop where operators differ in capability and fatigue, to compare task-allocation strategies on performance and fairness.

- Showed that a single Markov chain is intractable (over a billion states with only 5 products) and that fatigue breaks the Markov property.
- Designed a hybrid model: M/M/1 arrival queue with a supervisor agent, bounded fatigue law, stochastic Petri nets and a multi-agent architecture.
- Defined the performance indicators and four allocation strategies; the project continued with numerical simulations and an AnyLogic model.

Statistics Project — Uncertain Catapult: Robust Inverse Design **Dec 2025**
Team of 5 — Design of experiments and robust optimisation, in charge of data processing
Objective : Find the settings that make a noisy catapult hit a target range of 1.95 m ± 0.10 m.

- Validated the measurement system with a gauge R&R study (ANOVA): repeatability and reproducibility below 0.03% of total variability.
- Modelled the range with full factorial, fractional and Taguchi designs (L25 model: $R^2 = 0.97$).
- Solved the inverse problem with a robust Nelder–Mead optimisation: 25/25 test shots within tolerance; compared with kriging + genetic algorithm.

Programming Project — Automated Warehouse Simulation **Spring 2025**
Team of 2 — Object-oriented programming in Python, in charge of the simulation code
Objective : Simulate an automated warehouse where robots, ejectors and a conveyor move products to the exit under a central coordinator.

- Designed the object model: one class per entity, a Robot base class with two specialised subclasses, and a coordinator that validates every action.
- Implemented target assignment (nearest free target, Manhattan distance) and breadth-first-search path finding with collision avoidance.
- Loaded configurations from XML; produced a movement log, a usage heatmap and statistics (cycles, processing time, blockages).

Innovation Project — Modular Electric Motorcycle **Oct 2024 – Jan 2025**
Team of 3 — PEIT, Entrepreneurship & Technological Innovation track
Objective : Identify an innovation that would help the electric motorcycle segment take off.

- Carried out the state-of-the-art study and wrote the final report.
- Linked 10 uses, 23 usage constraints and 20 technical characteristics through weighted cross-matrices calibrated on market benchmarks.
- Proposed six modular systems; the five main modules cover over 90% of the needs identified.

TIPE — Markov Chains: Queues & Monopoly **Sep 2023 – Jun 2024**
Individual research project — Lycée Lakanal
Objective : Use Markov processes to size queueing systems, then model the game of Monopoly.

- Derived the M/M/1 steady state and performance indicators; validated them with a Python simulator (10 s in theory, 9.998 s simulated).
- Showed that pooling two queues into one shared M/M/2 queue cuts the mean time in the system by 24%.
- Modelled Monopoly as a 120-state Markov chain to compute visit probabilities and payback times, showing three houses is the optimum.

PROFESSIONAL EXPERIENCE

Chantiers Modernes Construction (VINCI) — Paris, France **Jun 2025 – Jul 2025**
Construction Site Worker Intern — Formwork

- On-site immersion on a major infrastructure project (EOLE / Line 15 interfaces).
- Full prepare–formwork–pour–strip cycle for reinforced concrete: fibre-reinforced segments, walls, columns and a foundation raft in a shaft.
- Read technical drawings, took measurements with surveyor support and ran visual quality checks.
- Coordinated with waterproofing and rebar teams on sequencing and site readiness.

Tutoring London — Remote **Oct 2024 – Present**
Private Tutor

- Prepare high-school students for the scientific baccalauréat (mathematics and physics).

SKILLS AND INTERESTS

- Languages: **French (Native), English (C1), Spanish (B1)**, Chinese (Beginner)
- Modelling: Markov chains, queueing theory, stochastic Petri nets, multi-agent simulation, design of experiments (Taguchi), ANOVA, robust optimisation, kriging, genetic algorithms
- Software: Python (object-oriented programming, Tkinter), AnyLogic, Excel, LaTeX, Microsoft Office
- Sports and Leisure: **Tennis** (competitive, ranked 15/1), **Football** (5 years of competition), **Scuba diving** (PADI certified)
- Readings: philosophy and current affairs